Machine Learning

03-04-26

- we need to minimize mse by finding optimal values of w and b

$$y = wx + b$$

Loss function:

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

eg.

x	y
$x_1 = 1$	$y_1 = 2$
$x_2 = 2$	$y_2 = 3$

$$mse = \frac{1}{2} \sum_{i=1}^{2} (y_i - \hat{y}_i)^2$$

with hypothesis function : $y = wx + b$

$$loss = mse = L = \frac{1}{2} \sum_{i=1}^{2} [y_i - (wx_i + b)]^2$$

$$L = \frac{1}{2} \left\{ [y_1 - (wx_1 + b)]^2 + [y_2 - (wx_2 + b)]^2 \right\}$$

$$= \frac{1}{2} \left\{ [2 - (w \times 1 + b)]^2 + [3 - (w \times 2 + b)]^2 \right\}$$

$$L = \frac{1}{2}\left\{\underbrace{[2-(w\times 1+b)]^2}_{(a-b)^2 = a^2+b^2-2ab} + [3-(w\times 2+b)]^2\right\}$$

$$= \frac{1}{2}\left\{[2^2+(w+b)^2-2\times 2\times(w+b)] + [3^2+(2w+b)^2-2\times 3\times(2w+b)]\right\}$$

$$= \frac{1}{2}\left[4+w^2+b^2+2wb-4w-4b+9+4w^2+b^2+4wb-12w-6b\right]$$

$$L = \frac{1}{2}\left[13+5w^2+2b^2+6wb-16w-10b\right]$$

- At the start of training, we have randomly initialised w and b

if we put $w=1$ in above eqn

$$L = \frac{1}{2}\left[13+5\times 1+2b^2+6\times 1\times b-16\times 1-10b\right]$$

$$L = \frac{1}{2}\left[2+2b^2-4b\right]$$

$$\boxed{L = b^2-2b+1} \longrightarrow L = f(b)$$

if we put $b=0$ in above eqn

$$L = \frac{1}{2}\left[13+5w^2+2\times 0+6w\times 0-16w-10\times 0\right]$$

$$\boxed{L = 1/2\left[13+5w^2-16w\right]} \rightarrow L = f(w)$$

rel^n between L and w

$L = \frac{1}{2}[13 + 5w^2 - 16w]$

$\downarrow$

change w to minimize L

rel^n between L and b

$L = b^2 - 2b + 1$

$\downarrow$

change b to minimize L

$L = f(w) = \underline{g}w^2 + \underline{h}w + \underline{i}$

$g = \frac{5}{2}$ $h = -8$ $i = \frac{13}{2}$

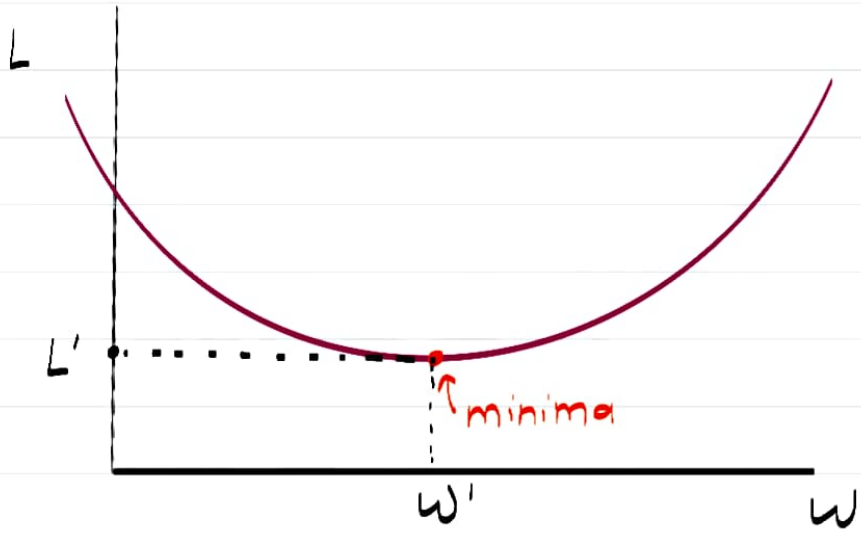
$L = f(b) = pb^2 + qb + r$

$p = 1$ $q = -2$ $r = 1$

both of these are quadratic equations

rel^n beth L and w
rel^n beth L and b } quadratic rel^n

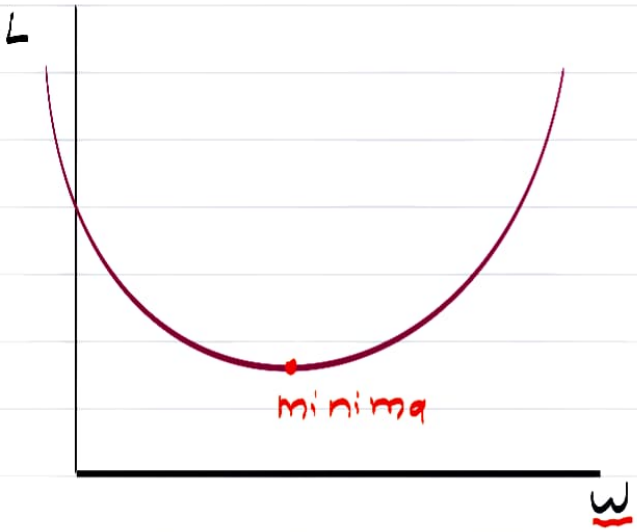
- quadratic rel^n creates <u>convex function</u>

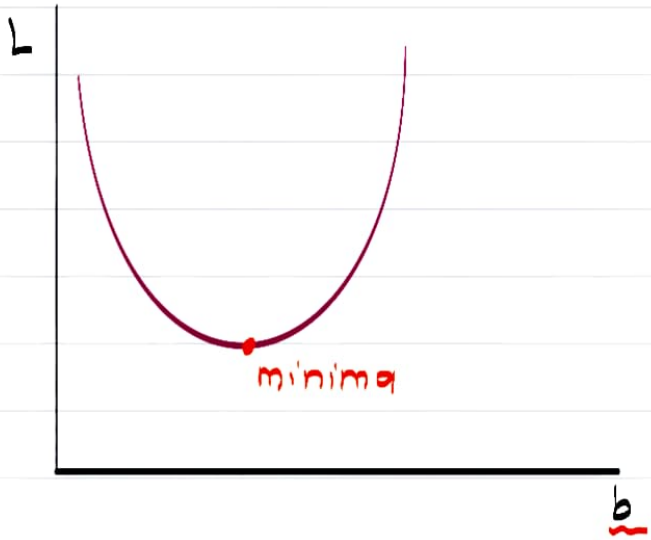


this type of curve is called convex function

convex function has a minima, which is a point on the curve where function value is minimum.

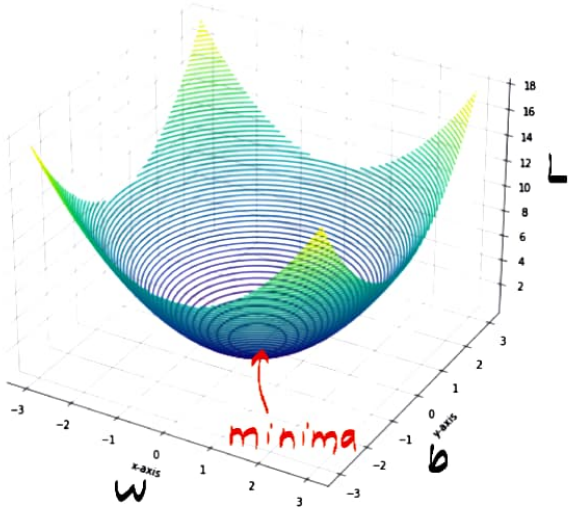
at $w = w'$ $\longrightarrow$ L is minimum i.e. loss is minimum





when we merge these two relⁿ together we get a contour plot

$$L = gw^2 + hw + pb^2 + qb - r$$



L = minima at $w = w'$ $b = b'$

minimizing loss L $\Rightarrow$ finding minima of this curve by finding w' and b'

Gradient descent is used to find minima of any given convex function

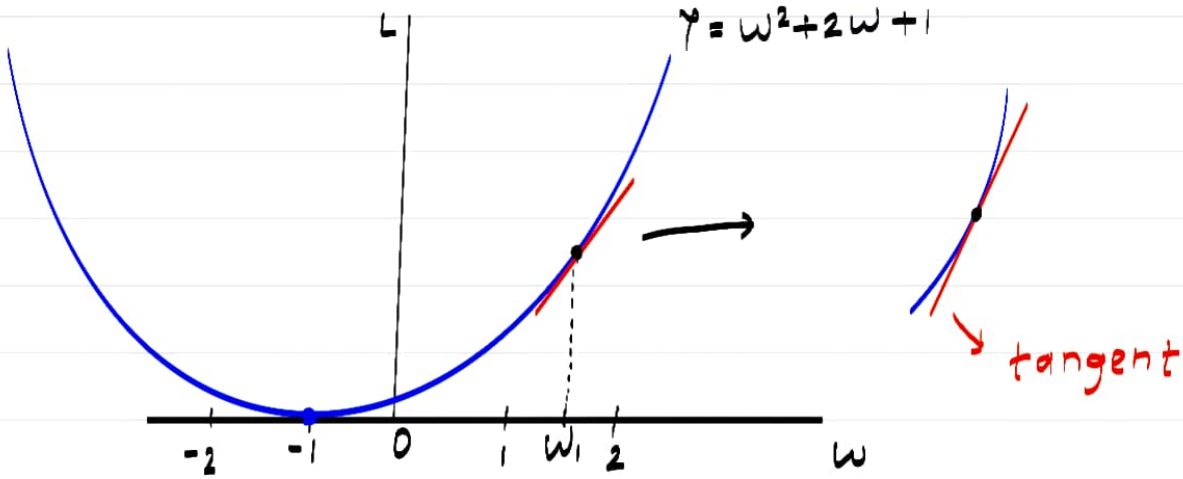
Gradient Descent

- uses slope / derivative of convex function to find its minima

eg.

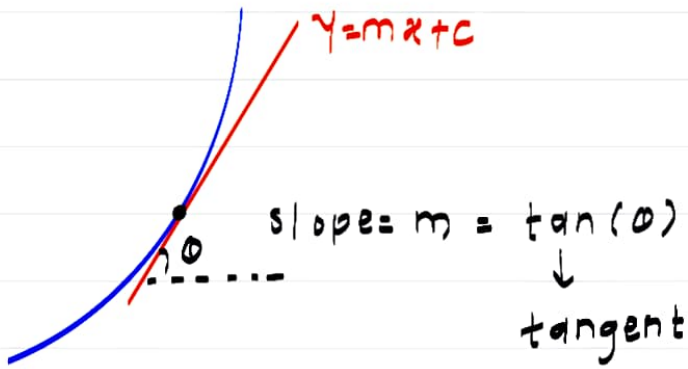
$L = w^2 + 2w + 1$

$L = (w+1)^2$



tanget - a straight line which touches the curve at a single point

slope of tanget = slope of the curve at point of contact



slope=m value at any given point on curve can be easily found using derivative function

Derivatives

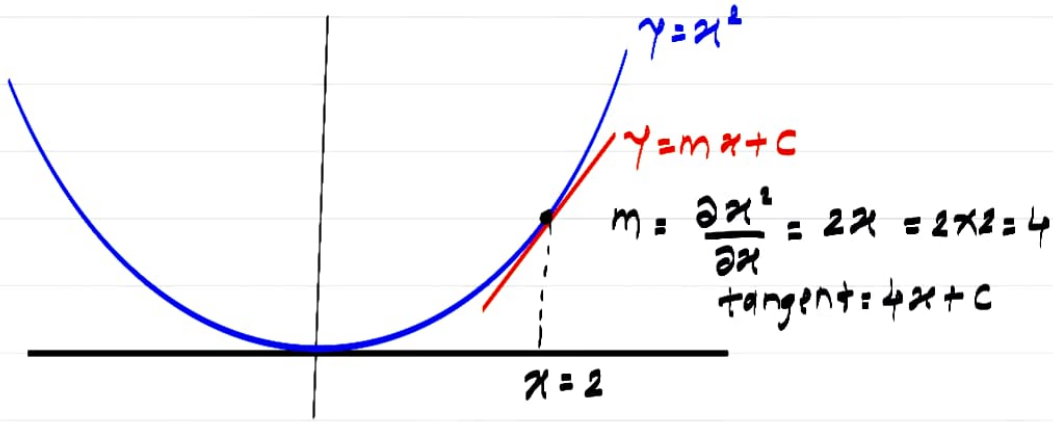
$y = f(x)$ → value of function at given x

$y' = \frac{\partial y}{\partial x} = \frac{dy}{dx}$ → value of slope of the function at given x

eg.

$y = x^2$

$\frac{\partial y}{\partial x} = 2x$



if $y = x^n$ then $\frac{\partial y}{\partial x} = n x^{n-1}$

$y = x^4$

$\frac{\partial y}{\partial x} = 4x^{4-1} = 4x^3$

$y = x^3$

$\frac{\partial y}{\partial x} = 3x^2$

$y = x^{-2}$

$\frac{\partial y}{\partial x} = -2x^{-3}$

$y = x$

$\frac{\partial y}{\partial x} = 1x^0 = 1$

$y = 5x$

$\frac{\partial y}{\partial x} = 5$

$y = 2x^2$

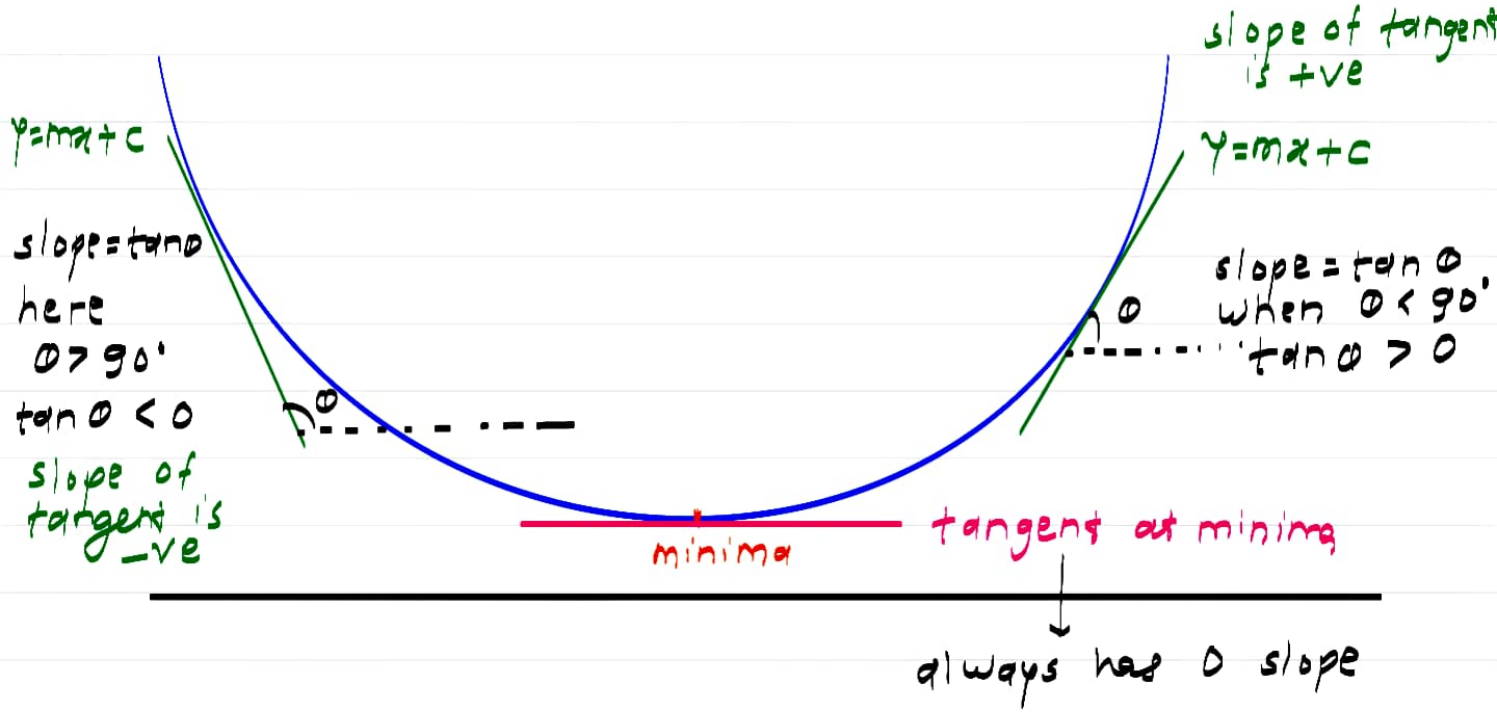
$\frac{\partial y}{\partial x} = 4x$

$y = 5x^2 + 3x + 8$

$\frac{\partial y}{\partial x} = 10x + 3 + 0$

$y = 10$

$\frac{\partial y}{\partial x} = 0$



slope of tangent

at minima → zero

right side of minima → +ve

left side of minima → -ve

Chain rule of derivatives

- simplifies computation of derivatives

$$y = [5-(2x^2+4x)]^2 \quad \text{find } \frac{dy}{dx}\ ?$$

$$y = 5^2 + (2x^2+4x)^2 - 2\times 5\times(2x^2+4x)$$
$$= 25 + 4x^4 + 16x^2 + 16x^3 - 20x^2 - 40x$$

$$\frac{\partial y}{\partial x} = \frac{\partial 25}{\partial x} + \frac{\partial 4x^4}{\partial x} + \frac{\partial 16x^2}{\partial x} + \frac{\partial 16x^3}{\partial x} - \frac{\partial 20x^2}{\partial x} - \frac{\partial 40x}{\partial x}$$

$$\frac{\partial y}{\partial x} = 0 + 16x^3 + 32x + 48x^2 - 40x - 40$$
$$\frac{\partial y}{\partial x} = 16x^3 + 48x^2 - 8x - 40$$

$$\boxed{\frac{\partial y}{\partial x} = 16x^3 + 48x^2 - 8x - 40}$$

using chain rule

$$y = [5 - (2x^2 + 4x)]^2$$

lets put $z = 5 - (2x^2 + 4x)$

$$z = 5 - 2x^2 - 4x$$

$$y = z^2 \qquad z = 5 - 2x^2 - 4x$$

$$\underbrace{\frac{\partial y}{\partial z} = 2z \qquad \frac{\partial z}{\partial x} = 0 - 4x - 4 = -4x - 4}$$

$$\frac{\partial y}{\partial x} = \frac{\partial y}{\cancel{\partial z}} \times \frac{\cancel{\partial z}}{\partial x} = \frac{\partial y}{\partial x}$$

$$= \frac{\partial y}{\partial z} \times \frac{\partial z}{\partial x}$$

$$= 2z \times (-4x - 4)$$

$$= 2(5 - 2x^2 - 4x) \times (-4x - 4)$$

$$= -8(5 - 2x^2 - 4x) \times (x + 1)$$

$$= -40x + 16x^3 + 32x^2 - 40 + 16x^2 + 32x$$

$$\boxed{\frac{\partial y}{\partial x} = 16x^3 + 48x^2 - 8x - 40}$$